

Felipe Hidalgo

Pipeline TD | Senior Software Engineer | Rendering Engineer

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Location: Bournemouth, UK | **Email:** fi.hq@hotmail.com | **Phone:** +44 7543 688502

LinkedIn: [linkedin.com/in/fhidargue](https://www.linkedin.com/in/fhidargue) | **Github:** github.com/fhidargue | **Showreel:** [youtube.com/watch?v= JaxgaLM55I](https://www.youtube.com/watch?v=JaxgaLM55I)

Summary

Software Engineer specializing in tools, pipeline development, and high-performance systems, with experience building C++ and Python applications for animation, VFX, graphics, and real-time content. My background combines professional software engineering with hands-on work in computer animation and graphics, including rendering, simulation, workflow automation, and performance optimization. I have developed cross-DCC workflows across Maya, Houdini, Unreal Engine 5, and OpenUSD, covering asset publishing, validation, animation retargeting, motion capture cleanup, and automated data pipelines. I enjoy solving complex technical problems and building reliable, scalable tools and systems that improve workflows and help technical and creative teams work more effectively.

Skills

Programming: C++, Python, Java, JavaScript, TypeScript, Go, C#, .Net

Pipeline & DCC: OpenUSD, Maya, OpenMaya API, Houdini, UE5, Pipeline Development, Workflow Automation, Asset Publishing, Validation

Tools & Platforms: PySide6, PyQt, Git, Linux, Docker, Podman, AWS, Azure, Google Cloud

Graphics & Simulation: OpenGL, GLSL, Real-Time Rendering, Path Tracing, Ray Tracing, Embree, Rendering Optimisation, Linear Algebra, Vector Mathematics

Software Engineering: Software Architecture, Data Structures & Algorithms, Debugging, Testing, APIs, Distributed Systems, SQL, MongoDB, Multithreading (TBB), CI/CD

Languages: Spanish, English, Italian

Technical Projects

Adaptive Scheduling for Coherent Wavefront Rendering (C++17, Python, OpenUSD) | 2026 | [Github](#) | [Video](#)

- ◆ Built a CPU wavefront path tracer using OpenUSD scene description to test whether adaptive scheduling improves rendering coherence and throughput.
- ◆ Achieved a 10% improvement in shading performance and up to 99% memory locality through adaptive scheduling and coherent workload grouping.
- ◆ Implemented core rendering algorithms and data structures (NEE, MIS, adaptive sampling, data-oriented design, cost-aware Russian Roulette), validated through TDD and A/B testing.
- ◆ Resolved cross-platform inconsistencies to support consistent builds and behaviour across macOS and Linux.
- ◆ Automated test data generation with Python scripts to drive large-scale render benchmarking.
- ◆ Built a PySide6 UI with OpenGL rendering for real-time render visibility, supporting artist-facing workflows.

USD Translation Pipeline (Maya → Unreal Engine 5) | 2026 | [Github](#) | [Video](#)

- ◆ Developed a validation-driven publishing pipeline that automated Maya-to-Unreal Engine OpenUSD workflows.
- ◆ Reduced scene transfer time from ~30 minutes to ~10 seconds by automating export, validation, and ingestion.
- ◆ Developed a SceneGraph abstraction to standardize Maya DAG data for reliable cross-DCC interoperability.
- ◆ Built configurable validation rules to catch invalid assets before they entered downstream production workflows.
- ◆ Developed pluggable registry backends using SQLite and MongoDB for traceable and reproducible asset publishing.
- ◆ Integrated Maya and OpenUSD into a modular pipeline supporting automated Unreal Engine 5 ingestion.
- ◆ Designed reusable, extensible components that made it easier to support additional workflows and integrate new DCC tools.

Motion Capture Retargeting & Cleanup Pipeline (Unreal Engine 5 → Houdini) | 2026 | [Eternally Dancing](#)

- ◆ Built a cross-DCC animation workflow supporting collaboration between technical and animation stages, including retargeting, cleanup, and procedural refinement.
- ◆ Reduced manual mocap cleanup effort by ~70–85% by shifting appropriate processing from Houdini to UE.
- ◆ Applied reference-driven animation corrections to improve motion quality, realism, and consistency.
- ◆ Integrated Unreal Engine and Houdini workflows to support efficient animation cleanup, refinement, and final polish.
- ◆ Reduced animation iteration time by restructuring workflows and automating stages across both DCCs.
- ◆ Built a workflow supporting collaboration across retargeting, animation cleanup, and procedural animation stages.

Real-Time Hair Simulation System (Python + OpenGL) | 2025 | [Github](#)

- ◆ Developed a physically based hair simulation using Verlet integration and constraint solvers, enabling stable real-time strand dynamics.
- ◆ Increased visual hair density by ~10× using GPU-driven follower strands without increasing simulation cost.
- ◆ Implemented a GLSL ribbon rendering pipeline for efficient real-time visualization of simulated hair.
- ◆ Developed interactive real-time parameter controls that accelerated simulation iteration and improved usability.

Snow Globe Simulation Engine (TypeScript, Three.js + WebGL) | 2026 | [Github](#) | [Video](#) | [Live Demo](#)

- ◆ Developed a real-time particle simulation system using Position-Based Dynamics (PBD) and Verlet integration for stable snow and marble dynamics.
- ◆ Reduced collision checks by over 90% using spatial hashing, enabling scalable real-time particle simulation.
- ◆ Implemented collision and constraint systems that maintained stable particle interactions during simulation.
- ◆ Rendered up to 20,000 particles in real time using GPU instancing.
- ◆ Designed a modular simulation architecture supporting scalable particle dynamics, collision handling, and rendering systems.

Experience

Senior Software Developer

Konrad Group, San José, Costa Rica | September 2021 – October 2025

- ◆ Built Python automation tools for translation, content, and email workflows, reducing manual specialist workload.
- ◆ Modernized data pipelines using Java, Spring Boot, real-time APIs, and database integrations, cutting turnaround from 4 hours to 10 minutes.
- ◆ Developed full-stack applications using React, Next.js, Vue.js, JavaScript, and TypeScript for large-scale client projects.
- ◆ Resolved critical build, dependency, and environment issues across Node.js, AEM, ACS, Magento 2, third-party libraries, and production environments during high-pressure deliveries.
- ◆ Designed real-time systems using Azure Queues, Stripe APIs, and cloud services, supporting thousands of concurrent users without disrupting purchases.
- ◆ Built reusable validation frameworks, automated testing, and CI/CD workflows using AWS, Google Cloud, Azure, Docker, and Podman, improving reliability and delivery.
- ◆ Served as a Subject Matter Expert (SME), leading architecture reviews and proofs-of-concept for scalable APIs, databases, authentication, and distributed systems.
- ◆ Owned end-to-end delivery of complex software solutions, collaborating with cross-functional teams to translate business requirements into scalable, maintainable systems deployed to production.
- ◆ Mentored developers through architecture workshops, pair programming, and technical documentation, improving team capability and long-term maintainability.

Education

MSc Computer Animation and Visual Effects

Bournemouth University – United Kingdom
2025 – 2026 (Expected August)

Focused on VFX pipelines, tool development, and data workflows, supporting real-time systems and production-facing automation.

Lic. Software Engineering

Universidad Fidélitas – Costa Rica
Graduated: March 2023

Studied software architecture, data structures, and systems development, with a focus on building scalable applications and engineering best practices.

Bachelor of Business Administration

Universidad de Costa Rica – Costa Rica
Graduated: December 2018

Focused on business operations, management, and decision-making, supporting a strong foundation in team collaboration.